Grade: Check Plus

STERN SCHOOL OF BUSINESS
NEW YORK UNIVERSITY

EXECUTIVE MBA
DC CLASS OF 2027

COR1.1205.85
Statistics and Data Analysis

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ASSIGNMENT 1

Attached Problem.

Due: February 24, 2026 before 6 PM

The probability distribution of the amount drawn from an ATM at a convenience store is tabulated below. In this table X = the amount withdrawn by a randomly selected customer and f(x) = P[X = x]. A customer can only withdraw the amounts indicated in the first column of the table below.

x	f(x)
10	0.10
20	0.15
40	0.30
60	0.15
80	0.25
100	0.05

✓Qa

a. What is the probability that a randomly selected customer withdraws at least $60 from the ATM? ✓Qb

b. Show by calculation that the expected value withdrawn from the ATM is $50 and that the standard deviation of the amount withdrawn from this ATM is $26.27. ✓Qc

c. The manager of this convenience score tells you that an expected value of $50 is not correct because it is impossible to withdraw $50 from the ATM. How would you answer the manager? ✓Qd

d. What is the probability that the amount withdrawn from the ATM will fall within one standard deviation of the mean? ✓Qe

e. What is the probability that the amount withdrawn from the ATM will fall within two standard deviations of the mean?

Assignment 1 - James Young

a. $f(x) = P[X = x]$

x	f(x)
10	0.1
20	0.15
40	0.30
60	0.15
80	0.25
100	0.05

$\left.\begin{matrix}0.15\\0.25\\0.05\end{matrix}\right\} 0.45$

The probability that a customer withdraws at least \$60 from the ATM is 45%, which is the sum of the probalities that a random customer will withdraw 60, 80 or 100.

$P(X \geq 60) = P(X=60) + P(X=80) + P(X=100)$

$= 0.15 + 0.25 + 0.05 = 0.45$

b. Show E of μ of x = 50 and standard deviation or $\sigma = 26.27$

x	f(x)	x(f(x))	x-50	$(x-50)^2$	$(x-50)^2 \times f(x)$
10	0.10	1	-40	1600	160
20	0.15	3	-30	900	135
40	0.30	12	-10	100	30
60	0.15	9	10	100	15
80	0.25	20	30	900	225
100	0.05	5	50	2500	125
		50			690

$\mu = \sum_{all\,x} (f(x)) = 1+3+12+9+20+5 = 50 \therefore \mu = 50$

$\sigma^2 = \text{variance} = \sum_{all\,x} (x-\mu)^2 = 160+135+30+15+225+125 = 690$

$\sigma = \sqrt{\sigma^2} = \sqrt{690} = 26.27 \therefore \sigma = 26.27$

c. The mean or expected value is not a prediction of the outcome of a particular withdrawal, rather it is a weighted average of all of the possible outcomes.

$\mu = 50$

$\sigma = 26.27$

d. probability that the amount withdrawn will fall w/in one standard deviation of the mean.

$P(\mu - \sigma \leq X \leq \mu + \sigma)$

$P(50 - 26.27) \leq X \leq 50 + 26.27$

$P(23.73) \leq X \leq P(76.27)$

Not continuous distribution so need to see which numbers fall in range:

X	f(X)
10	0.1
20	0.15
40	0.30
60	0.15
80	0.25
100	0.05

$P(X)$ w/in $1\sigma = 0.30 + 0.15 = 0.45$ or 45%.

e. same question but two standard deviations

$P(\mu - 2\sigma) \leq X \leq \mu + 2\mu) = 50 - 2 \cdot 26.27 \leq X \leq 50 + 2 \cdot 26.27$

$= -2.54 \leq X \leq 102.54$

from chart above we see that all numbers fall within range of -2.54 to $+102.54$, so the probability that X will fall within 2 standard deviations of the mean is 1.00 or 100%